Fiscal Year:	FY 2019	**Task Last Updated:**	FY 09/04/2018
PI Name:	Sekulic, Dusan D.Sc.		
Project Title:	BRAzing of Aluminum Alloys IN Space		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	40506	**Congressional District:**	6
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	11/04/2016	**End Date:**	11/03/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Rogers, Jan	**Contact Phone:**	256.544.1081
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	September 2018: CoPrincipal Investigators Dr. Fransaer and Dr. Seveno are no longer working on the project.		
COI Name (Institution):	Krivilev, Mikhail Ph.D. (Udmurt State University) Mesarovic, Sinisa Ph.D. (Washington State University)		
Grant/Contract No.:	NNX17AB52G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We propose to investigate the techniques of brazing (bonding by molten alloy) in microgravity for space construction and repair of spacecrafts damaged by micrometeoroids and orbital debris. A series of space (International Space Station - ISS) and terrestrial experiments are planned to be integrated with modeling and analysis. These are proposed to answer the practical and fundamental questions: (1) How can we take advantage of the absence of gravity and dominance of capillary forces in transporting the large quantity of molten braze? (2) What composition of the brazing alloy and the brazing technique will enable strong and efficient bonding of structural elements and repair? (3) Can this brazing be done using fluxless cladding? The phenomena to be studied under these conditions include molten metal kinetics, wetting and spreading of liquid clad/filler over topographically modified surfaces. The applied aspects include direct verification if brazing of Al alloys is transferable to orbit for rapid repair of the pressure wall. For the first time, brazing in space will be processed using the patented fluxless Trillium technology. Further development includes design and testing of a space-based brazing hot zone with the support from Marshall Space Flight Center (MSFC).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The impact of the proposed research on the life on Earth manifests itself through multifaceted benefits. These include: (i) uncovering so-far hidden physical/chemical/metallurgical aspects of metal bonding by brazing (extensively used in manufacturing) through uncoupling of the gravity impact, hence transforming the art of this technology to a predictive science for terrestrial applications, (ii) exploring a possibility to use novel materials as bonding alloys less sensitive on terrestrial hindrances (e.g., oxidation) in designing bonded assemblies, (iii) featuring sustainable resources use and less energy invested due to an expected broadening the range of applicability of bonding under adverse conditions, (iv) developing visualization of the brazing process in situ and in real time under vacuum and/or controlled atmosphere, (v) developing the kinetics of wetting theory by implementing phase field techniques that can be used to model capillary phenomena not only for liquid metals but for other material systems in both micro gravity and gravity (on Earth) conditions.
Task Progress:	This document constitutes the Year 2 report of the research work conducted within the scope of the NASA sponsored “BRazing Aluminum IN Space” (BRAINS) project (NNX17AB52G). The work has included both experimental and theoretical modeling segments. The participation of both collaborating teams (the University of Kentucky, UK, and the Washington State University, WSU), established in Year 1, continues. A full participation of the Russian team continues as well. Funding of the Belgian European Space Agency, Belgian Federal Science Policy Office's (ESA, BELSPO) and Russian teams (University of Leuven and Udmurt State University, respectively) was not established in Year 1 but ROSCOSMOS' support of the Russian team is under consideration. Both experimental and theoretical (modeling) work have been completed as planned at UK, WSU, and UdSU. The experimental work under terrestrial conditions was focused on defining and populating the map of parameters indicating domains of filling or not filling the micro/macro holes by surface tension driven liquid aluminum alloys. The orientation in the gravity field of the spreading fluid plane has been considered and an impact of gravity for larger alloy masses has been demonstrated. The modeling activities during the second year were focused on the following tasks/issues: (1) Modeling of surface tension controlled equilibrium formations of molten aluminum, (2) Modeling of the kinetics of the process of filling the holes by surface micro layers of liquid aluminum alloy. The experimental work in Year 2 explored both phenomenological and theoretical modeling for the hole-in-a-brazing-sheet filling problem in a controlled atmosphere brazing process. Solid a-Al formation during the brazing process promotes the possibility that liquid aluminum may travel over sharp edges through micro-channels. This may further support the phenomenon of changing liquid volume on the wall of the hole. It is established that the general outcome of filling vs. non-filling can be predicted by a theoretical model. The model has been developed using the principle of minimal potential energy. The model generally fits experimental results with some deviations, possibly due to assumptions made during the model development. Based on the prediction, “self-healing” (filling the hole) would be achieved if the hole diameter and depth of the brazing sheet have beforehand established appropriately based on suggested threshold values by the model. Oppositely, the liquid formation will be pinned around the hole instead of growing toward the center and filling the hole. As confirmed both experimentally and theoretically, it is favorable to have a deeper or narrower hole in order to fill an originally open hole on an aluminum brazing sheet. In Year 2 we have considered the impact of the orientation (an inclination of the wetted surface) in the gravity field. Gravity does have an effect on wetting of the liquid metal over AA3003. As the inclination angle gets smaller, the surface profile gets thinner, and the contact angle between the alumina substrate and the liquid metal becomes larger. Most importantly, the wetting distance is longer when the inclination of AA3003 is decreasing. It can also be concluded that the wetting distance becomes smaller and gravity effect seems to decrease when the quantity of liquid metal decreases. If a given quantity of liquid metal has a wetting distance on the vertical piece below 5.96 mm, it seems that gravity effect can be neglected. Theoretical analysis has included modeling equilibrium shapes of free surfaces of molten metal upon filling or non-filling the hole. Modeling was performed by using the Surface Evolver software. Agreement with empirical data has been satisfactory. The phase-field simulations of hole-filling in brazing reproduce the flow behavior similar to the experimentally observed one. The characteristic time of wetting is of the order of 1 second. The liquid flows in the hole and wets the back side of the substrate. This process is controlled by surface mobility of the triple line which significantly slows down the overall kinetics of wetting. Further study is planned to define precisely the surface mobility using the real-time videos of brazing. A transition to the axisymmetric formulation will be simulated in the model in order to get exact correspondence with the brazed geometry planned for the space experiment.
Bibliography Type:	Description: (Last Updated: 12/06/2025)
Abstracts for Journals and Proceedings	Yu C-N, Wu Y, Sekulic DP. "Capillary Flow on an Aluminum Brazing Sheet in a Gravity Field: A Phenomenological Study." International Brazing and Soldering Conference (IBSC), New Orleans, LA, USA, April 15-18, 2018. International Brazing and Soldering Conference (IBSC), New Orleans, LA, USA, April 15-18, 2018. , Apr-2018

Abstracts for Journals and Proceedings	Wu Y, Fu H, Hawksworth DK, Yu C-N, Sekulic DP. "Diffusion Fields in TRILLIUMTM Brazing Sheet under Different Heating Rates." International Brazing and Soldering Conference (IBSC), New Orleans, LA, USA, April 15-18, 2018. International Brazing and Soldering Conference (IBSC), New Orleans, LA, USA, April 15-18, 2018. , Apr-2018
Abstracts for Journals and Proceedings	Wu Y, Yu C.N. Sekulic DP. "Aluminum Braze Triple Line Movement in a Wetting/Non-wetting Setting under Variable Surface Roughness and Inclinations." International Brazing and Soldering Conference (IBSC), New Orleans, LA, USA, April 15-18, 2018. International Brazing and Soldering Conference (IBSC), New Orleans, LA, USA, April 15-18, 2018. , Apr-2018
Awards	Sekulic DP. (Dusan P. Sekulic) "Invited Lecture: The Role of Wetting in Joining, 14th International Ceramics Congress, Perugia, Italy, June 4-8, 2018." Jun-2018
Books/Book Chapters	Bar-Cohen Y, Sekulic DP, Pan R, Babu SS, Hu A, Bridges D, Bao X, Badescu M, Le HJ, Sherrit S. "Ch 12. Metal Joining Techniques Using Brazing." in "Advances in Manufacturing and Processing of Materials and Structures." Ed. Y. Bar-Cohen. Boca Raton, FL : CRC Press, Taylor & Francis Group, LLC, 2018. ISBN: 9781138035959. https://doi.org/10.1201/b22020 , Sep-2018
Books/Book Chapters	Pan R, Sekulic DP. "Ch 13. Diffusion Bonding." in "Advances in Manufacturing and Processing of Materials and Structures." Ed. Y. Bar-Cohen. Boca Raton, FL : CRC Press / Taylor & Francis Group, LLC, 2018. https://doi.org/10.1201/b22020 , Sep-2018